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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-48

AC Type-2 Charging - I

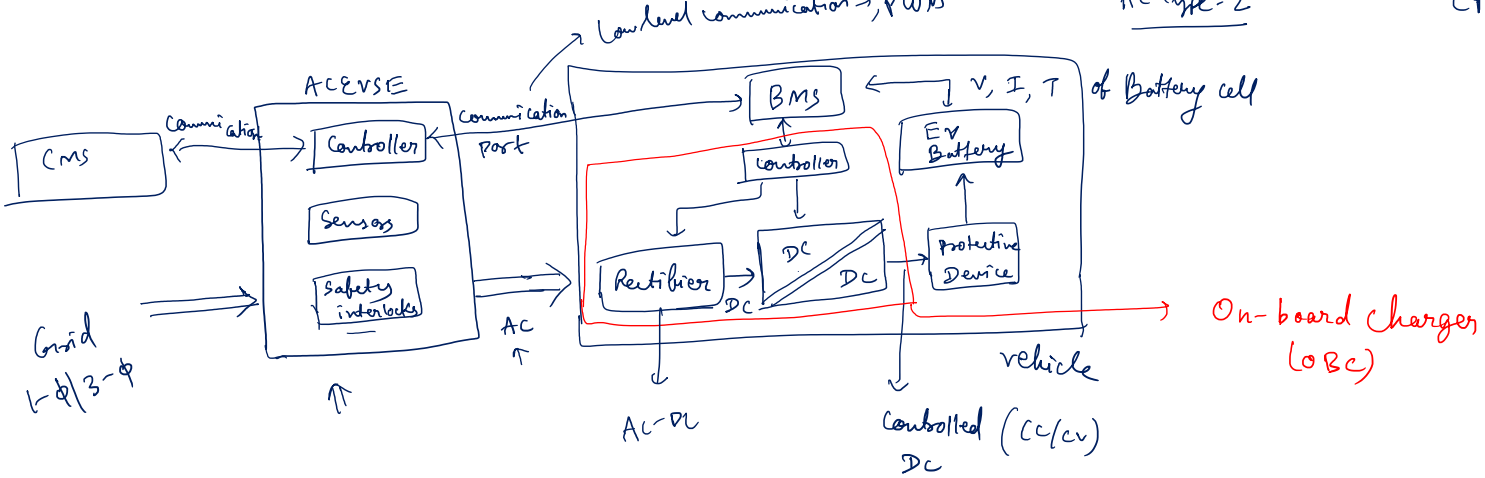
Dr. Apurv Kumar Yadav

Department of Electrical Engineering



Recap

- Power conversion unit : AC-DC, isolated DC-DC



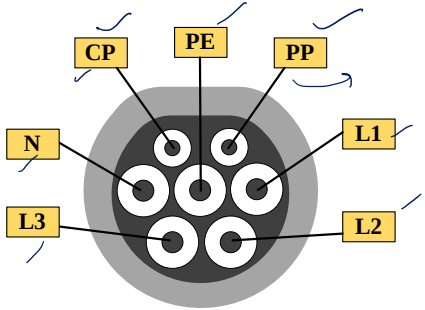
AC Type-2 Charger

- The communication → PWM communication over CP (Control Pilot) pin.
- EV recognizes plug-in via PP (proximity pilot)
- Vehicle receives information about maximum possible charging current and charger receives vehicle ready status
- Vehicle is the master

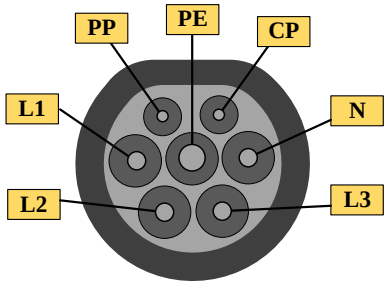


Type 2 AC **Charger Plug**

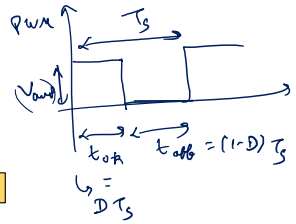
BMS in EV who is the master



Charger Plug
Type 2

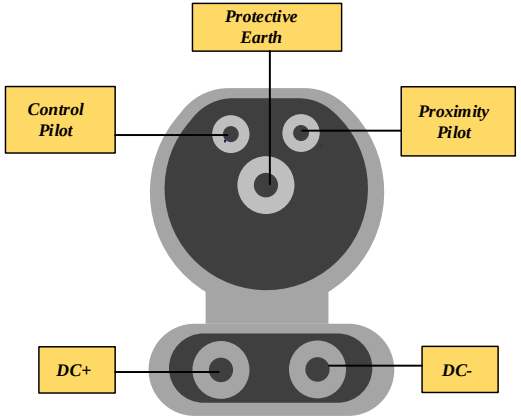


AC vehicle inlet
Type 2

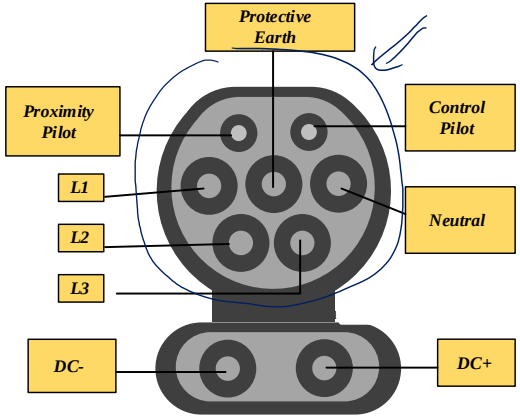


V_{amp} is changed
D is changed

DC CCS2



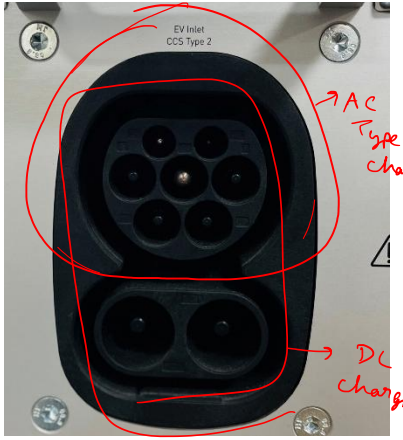
Charger Plug



Vehicle Inlet Connector



Charger Plug



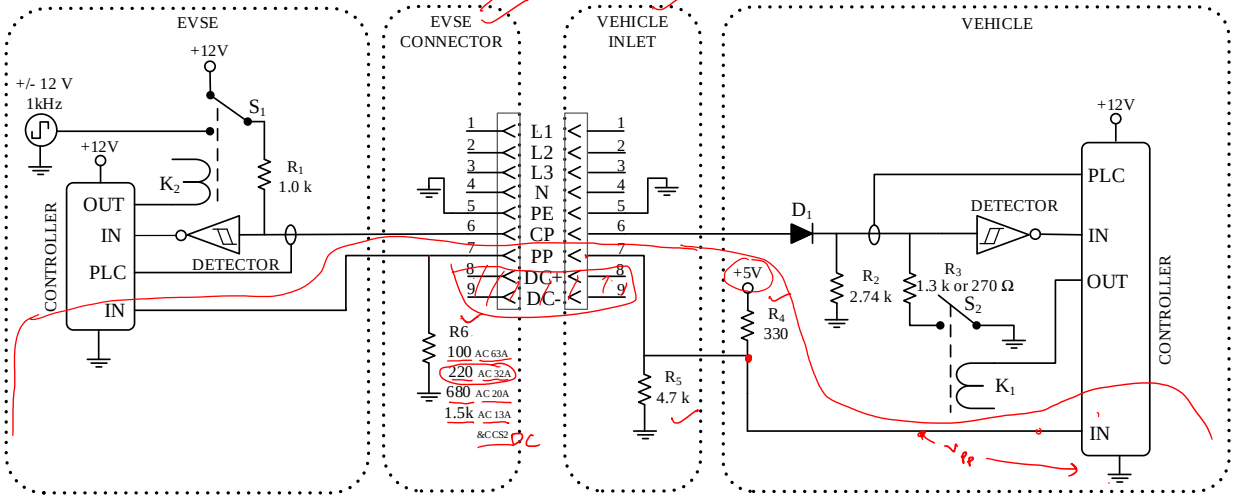
Vehicle Inlet Connector

→ AC charging is actually a slow rate of charging

→ DC charging can do charging at higher power level

AC Charging (Type-2 or CCS2): PP Circuit

- PP (Proximity Pilot) detected the adaptor (in the EV)
- The resistance R_6 value indicates (to EV) the maximum current rating of the plugged cable (from EVSE).



Bahrami, Ali. (2020). EV Charging Definitions, Modes, Levels, Communication Protocols and Applied Standards. 10.13140/RG.2.2.15844.53123/11.

AC Charging (Type-2 or CCS2): PP Circuit

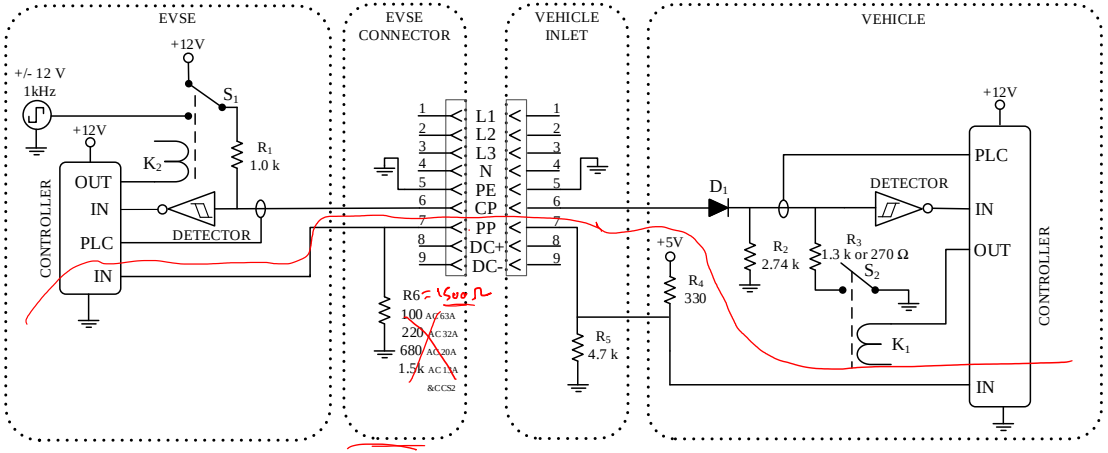
Proximity resistance circuitry tolerances in SAE J1772 and IEC 61851-1

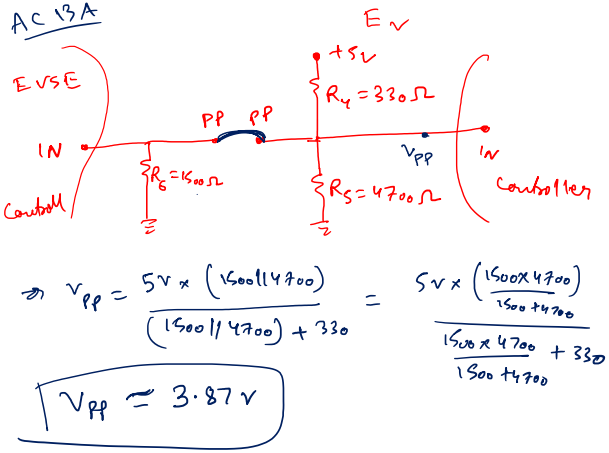
Resistance between PP-PE	Tolerance range	Max. Current	Conductor Size	Rated voltage on PP pin
> 4500 Ω	-	Error condition or disconnected plug	-	> 4.37V
1500 Ω	1100-2460	13A AC, or DC CCS2	1.5mm²	3.87V
680 Ω	400-936	20A AC	2.5mm²	3.21V
220 Ω	164-308	32A AC	6mm²	1.94V
100 Ω	80-140	63A(3-phase)	16mm²	1.14V
<60 Ω	-	Error condition	-	<0.76V

Bahrami, Ali. (2020). EV Charging Definitions, Modes, Levels, Communication Protocols and Applied Standards. 10.13140/RG.2.2.15844.53123/11.

AC Charging (Type-2 or CCS2): PP Circuit

- PP detected the adaptor (in the EV)





The details are in given in SAEJ1772 standards (in the form of SAE J1772 signalling protocol) and also adopted in IEC 61851

Thank You